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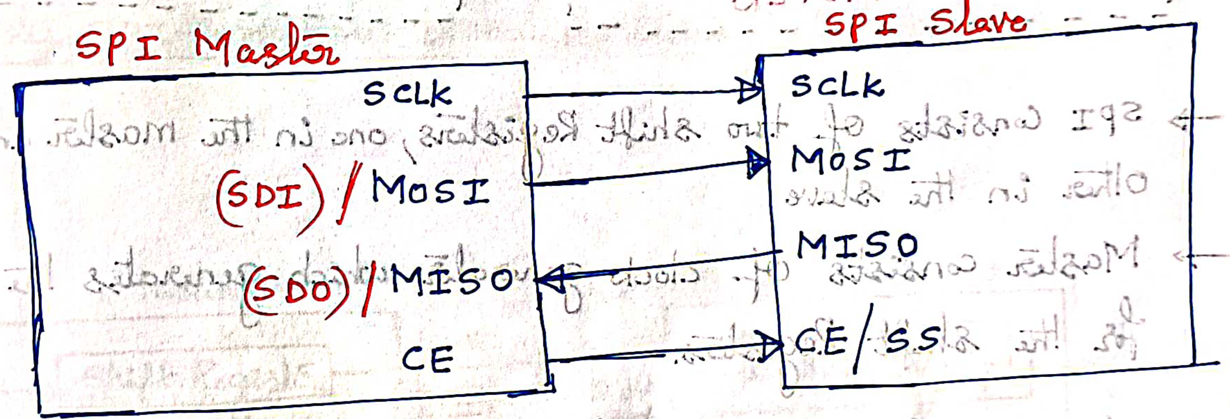
* Serial peripheral Interface : [SPI]

- Full - Duplex, serial communication protocol.
- Synchronous communication protocol.
- Four wire communication protocol
- Single - Master & Multi - slave
- Widely used for short - distance communication used in Embedded system.
- It is also called as "4 - Wire serial bus."

NOTE: Communication protocols:
- UART
- I2C
- SPI

Embedded System

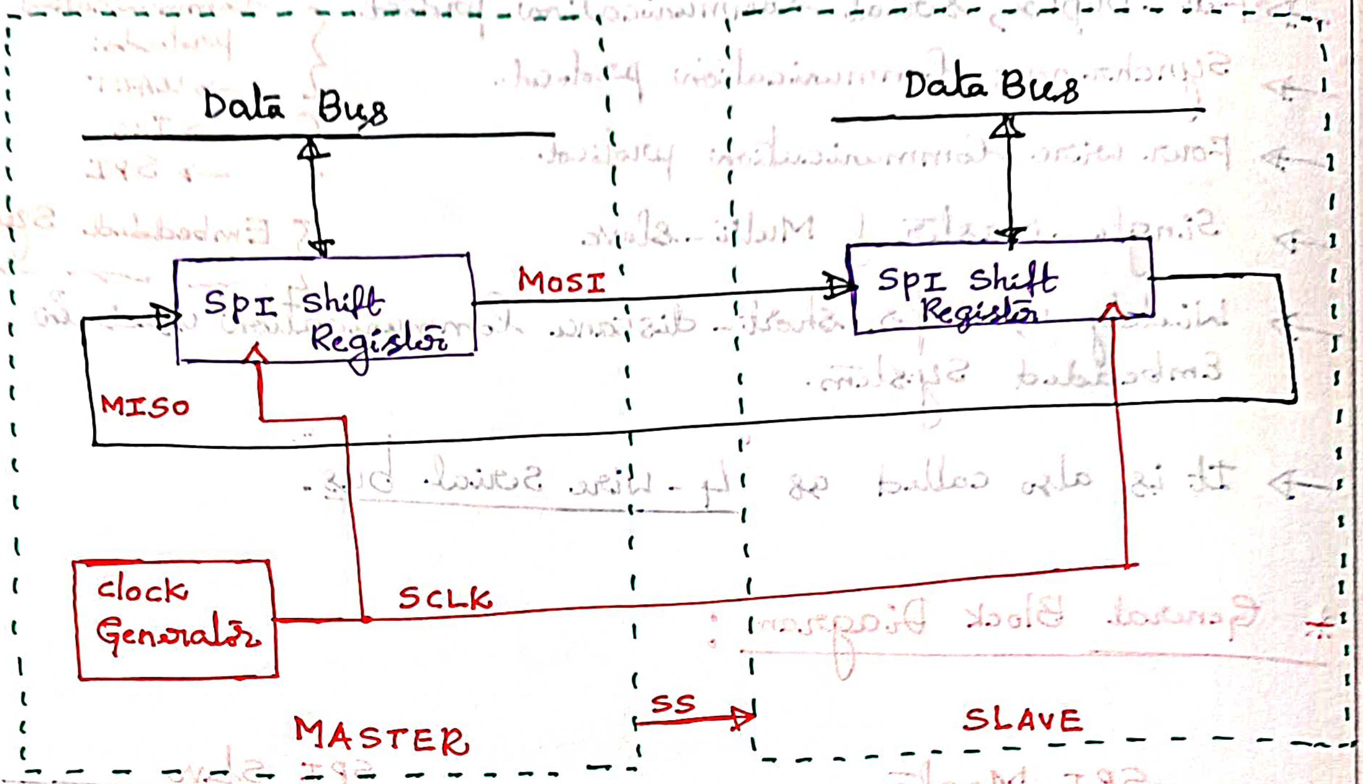
* General Block Diagram :



Pin Names :

- MOSI (Master out serial In) / SDI (Data In)
- MISO (Master In serial out) / SDO (Data out)
- SCLK (serial clock)
- CE (chip Enable) / SS (chip select)
- The above four pins make the SPI a 4 - wire Interface.
- The SPI devices use only 2 pins for data transfer called SDI (Data In) and SDO (Data out).

* SPI Architecture :



- SPI consists of two shift Registers, one in the master and the other in the slave
- Master consists of clock generator which generates the clock for the shift Registers.
- The clock input of the shift registers can be +ve or -ve edge triggered.
- The shift Registers are 8 bits long. It means that after 8 clock pulses, the contents of the two shift registers are interchanged.
- If the master wants to send a byte of data, it places the byte in its shift Register and generates 8 clock pulses. After 8 clock pulses, the byte is transmitted to the slave register
- If the master wants to receive a byte of data, the slave side should place the byte in its shift register and after 8 clock pulses, the data will be received by the master shift register.
- SPI is full duplex meaning that it may send and receive data at the same time.

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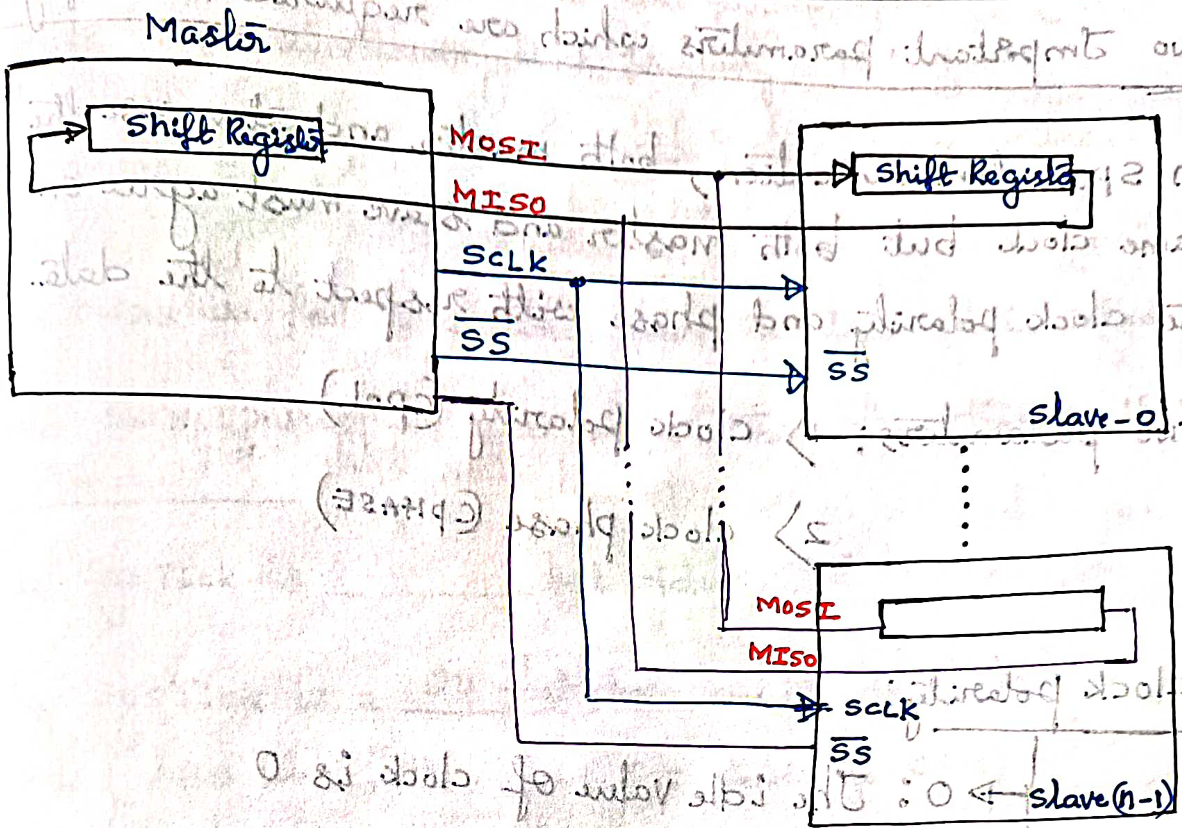
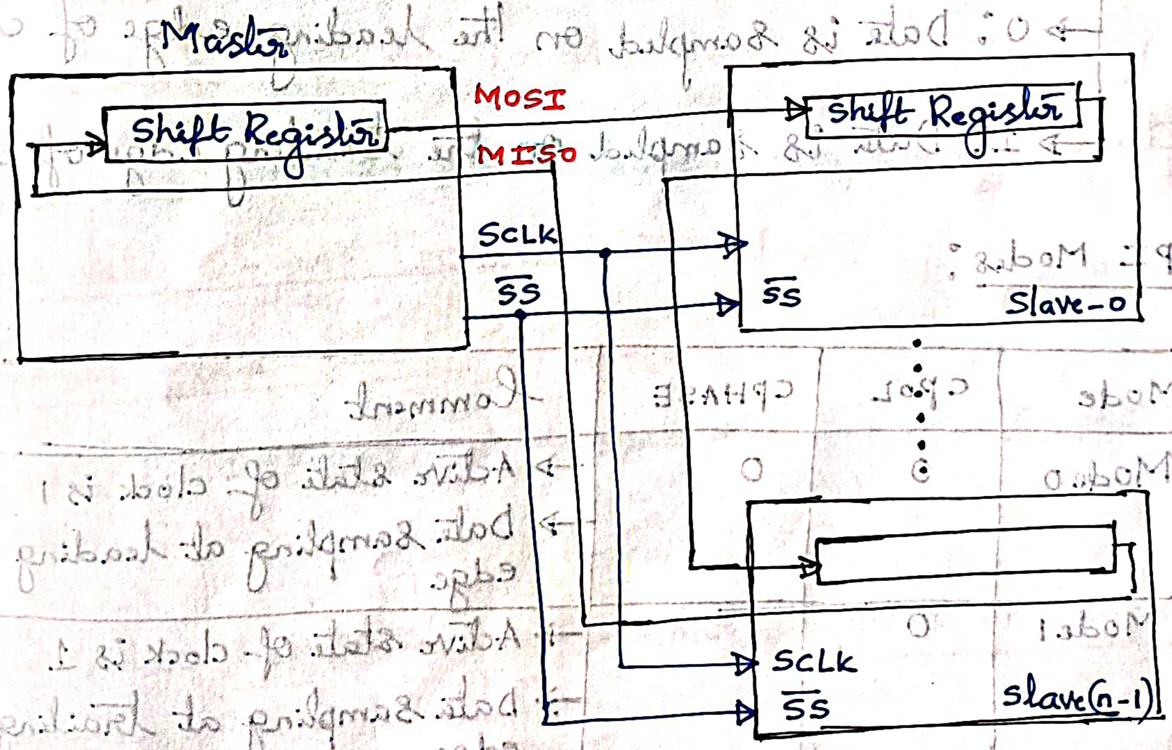


Master SPI connection to multiple slaves SPI in parallel



Master SPI connection to multiple slaves in serial

* Two Important parameters which are required to configure:

→ In SPI communication, both master and slave use the same clock but both master and slave must agree on the clock polarity and phase with respect to the data.

→ Two parameters: 1) clock polarity (CPOL)

2) clock phase (CPHASE)

→ clock polarity:

- → 0: The idle value of clock is 0
- → 1: The idle value of clock is 1

→ clock phase:

- → 0: Data is sampled on the leading edge of clock
- → 1: Data is sampled on the trailing edge of clock.

→ SPI Modes:

Mode	CPOL	CPHASE	Comment
Mode 0	0	0	→ Active state of clock is 1 → Data sampling at leading edge
Mode 1	0	1	→ Active state of clock is 1 → Data sampling at trailing edge
Mode 2	1	0	→ Active state of clock is 0 → Data sampling at leading edge
Mode 3	1	1	→ Active state of clock is 0 → Data sampling at trailing edge